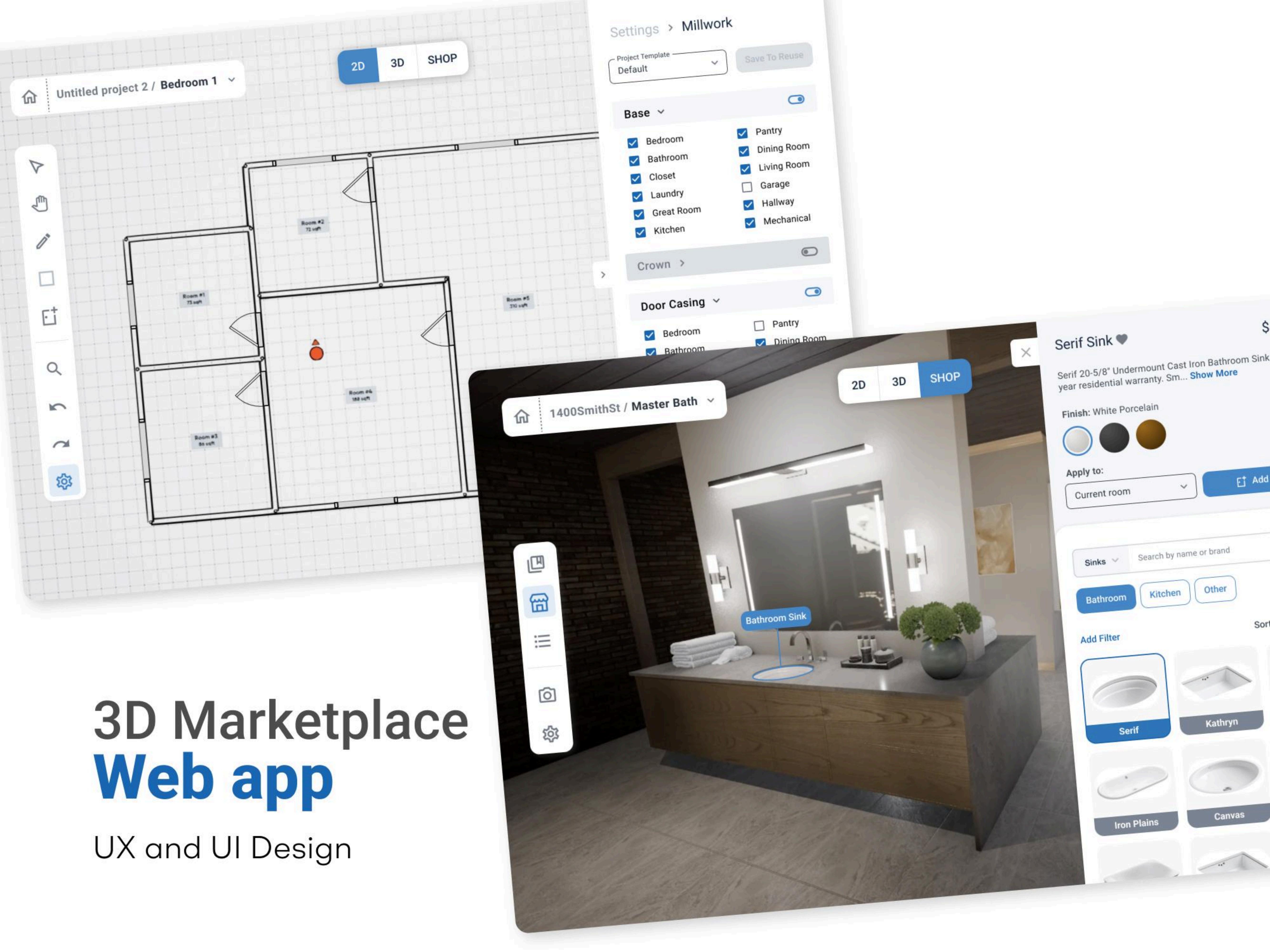




3D Marketplace Web app

UX and UI Design

😞 Problem

Builders and suppliers struggle with (1) a **lack of clarity** in design, pricing, and inventory status and (2) managing **multiple projects** with thousands of product options.

😐 Current solutions

The solutions offered on the market are either highly customizable but **expensive** and time consuming, or quick view options with **minimal customization**.

😎 VIRTUALhaus

VIRTUALhaus is the first **3D Marketplace** that connects building product suppliers directly with their customers. Turning 2D floor plans to 3D with full Customization & Walkability.

1

PROCESS

1.1 My role

I was a full-time "UX/UI team of one" at VIRTUALhaus from March 2022 to September 2022. My two main challenges were creating the UI design (and updating it) and experimenting with new features.

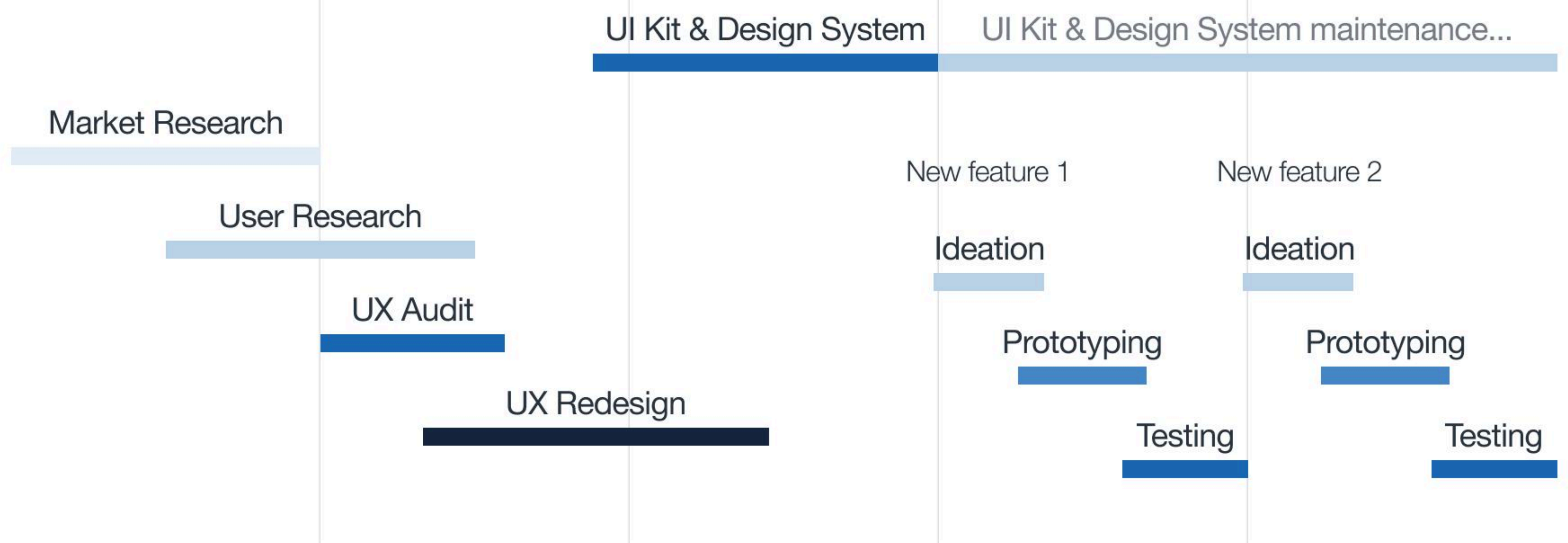
I was responsible for planning and executing the whole process: user interviews, user flows, prototypes, UI design, user testing, and presenting findings to the stakeholders.

1 month

2 months

3 months

4 months



When I first joined the company I focused on understanding the market and the users. To learn more about the product VIRTUALhaus had already developed, I did a user experience evaluation and redesigned some user flows and reorganizing the information architecture.

I then used an Ideate / Prototype / Test method to **explore new features** and **save development time**.

2 DISCOVERY

2.1 Understanding the problem

We discovered that shopping for building products today can be a 6+ months stressful process that requires a lot of planning and searching from both the

builder and the supplier.

6 months later

Builders have a hard time helping their clients envision the design



They drive around from showroom to showroom showing samples to their clients



Prices increase or change often, and that product their client chose? Out of stock!



They have to start all over again



That beautiful porcelain sink and the widespread faucet DO NOT fit together!

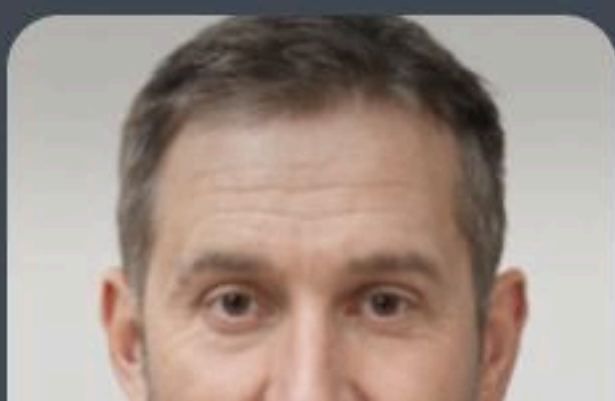


2.2 Emphathizing with our user

Being a Markeplace means we needed to connect two sides and give value to both. To be able to do this we first need to put ourselves in their shoes, for that we held 9 inerviews with both builders and suppliers.

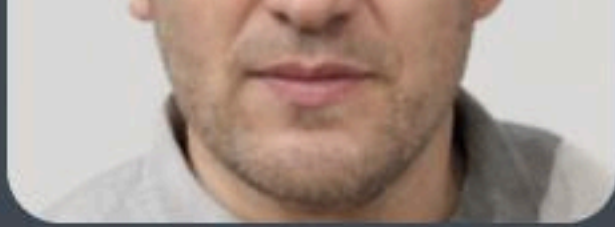
About

Construction Manager at his own small home building company, with over 15 years of experience. He's well know in the area, has a very busy agenda, not much experience with digital tools.



Goals

- Delegate or automate tasks effectively to help the company grow.
- Spend more time with his family.
- Travel abroad without losing clients.



Marco, 47
Construction Manager

AUSTIN, TX

Married, 2 kids.

Software & Apps

- Phone and email for communication.
- Microsoft Office.
- Facebook.

Frustrations

- He often feels overwhelmed with things to do at work.
- Doesn't want to continue working during weekends.
- Feels like he should be spending more time with his younger child.



Spencer, 36
Outside Sales Rep.

DALLAS, TX

About

Recently hired by a building materials manufacturer and supplier. He just moved to Dallas, but the company has 27 locations spread throughout the US. He is very active in social media and in his social circles.

In a relationship.

Software & Apps

- Email and text for communication.
- Microsoft Office, mostly Excel.
- Instagram, Snapchat, & YouTube.

Goals

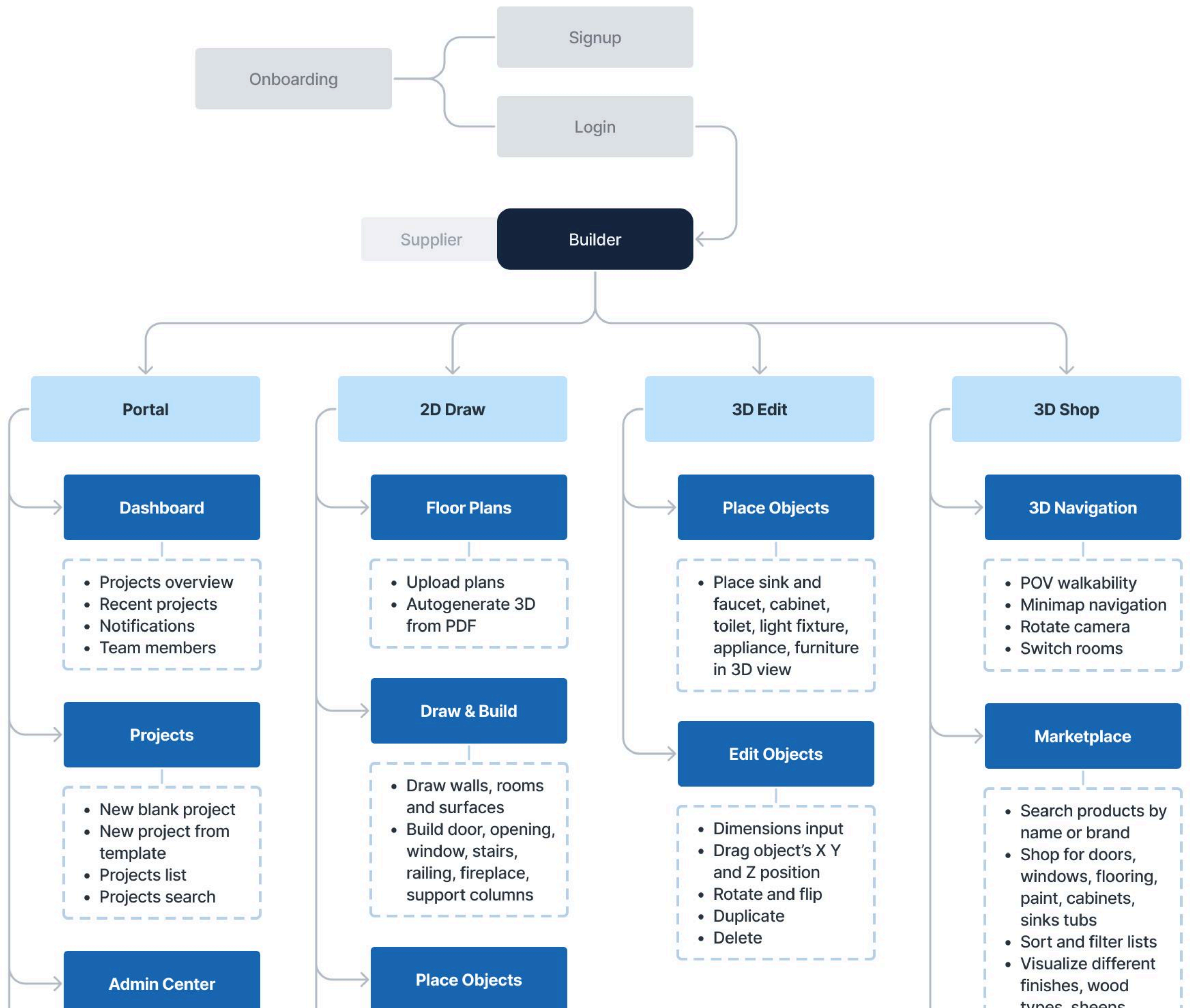
- Increase his income.
- Figure out his niche in the building industry.
- Expand his network connecting with more experienced people in construction.

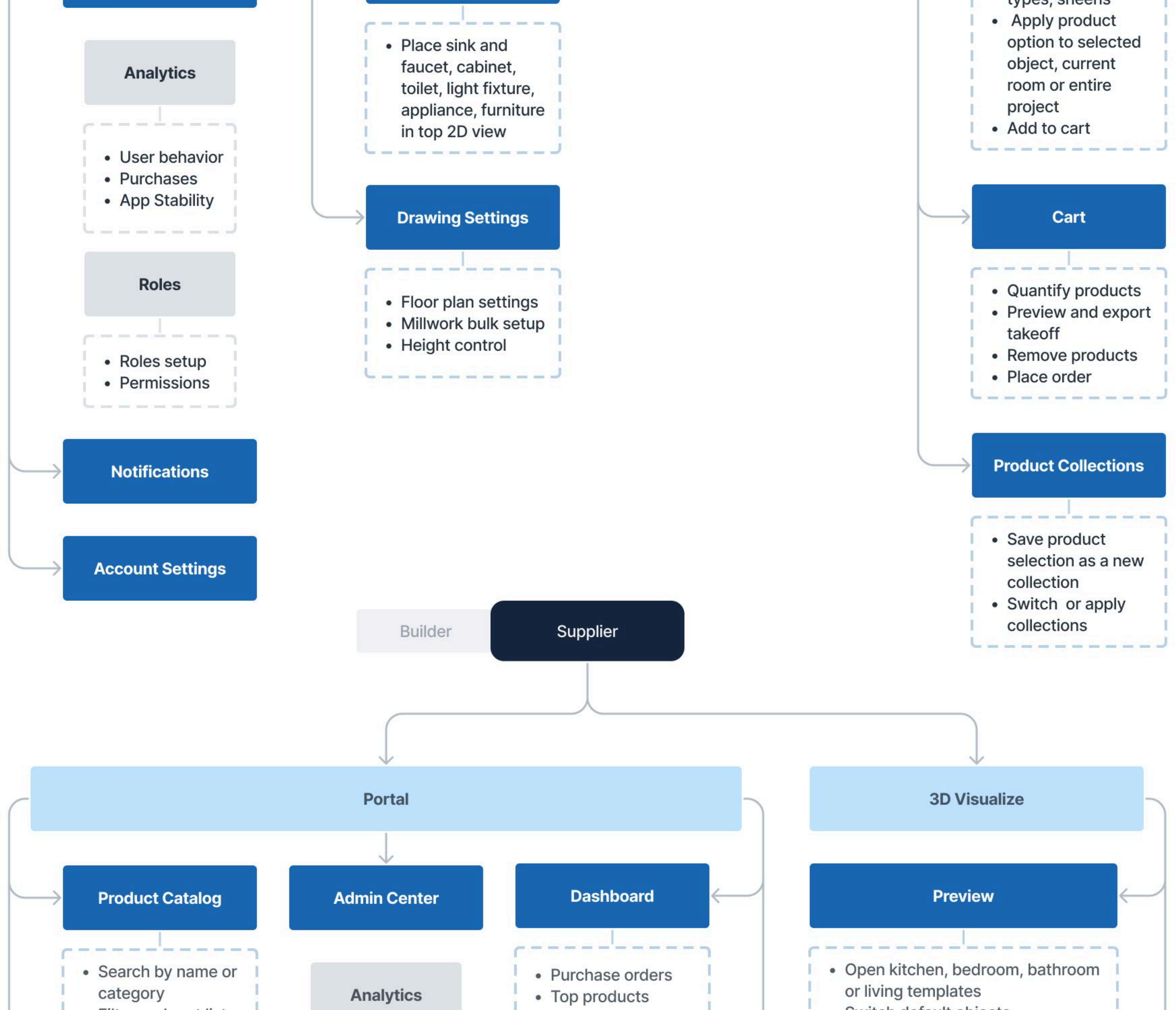
Frustrations

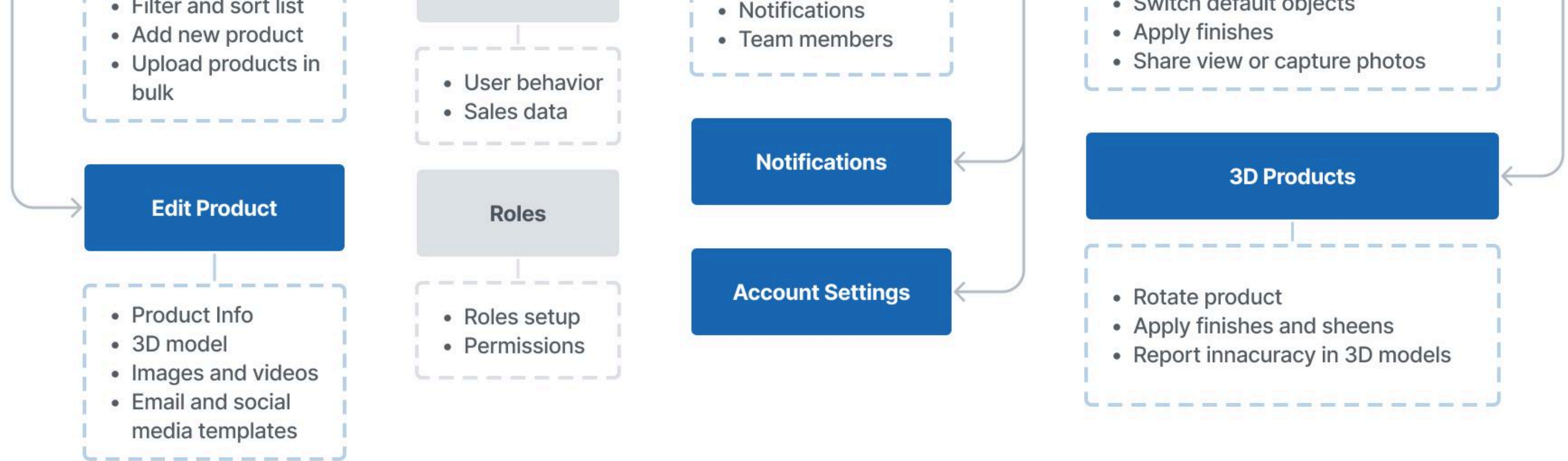
- Disappointed because he's not closing many sales.
- He doesn't know where to ask for help when he needs it.
- Often, he feels his efforts are fruitless.

3.1 Information Architecture

We knew all about the people involved and their problems, but now we were asking ourselves "how can we solve this for them?". Together the team defined the features our platform would have and I created the information architecture.



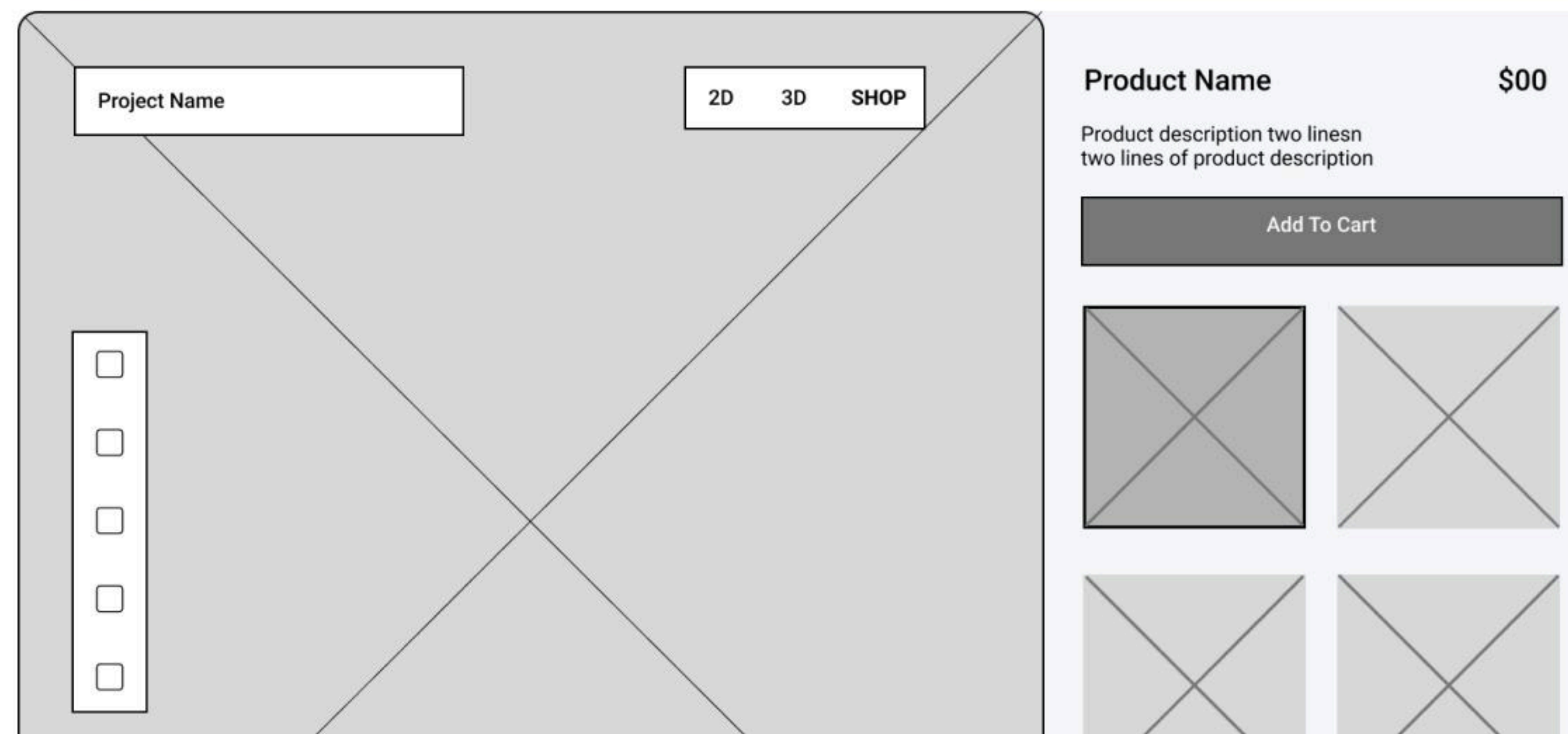


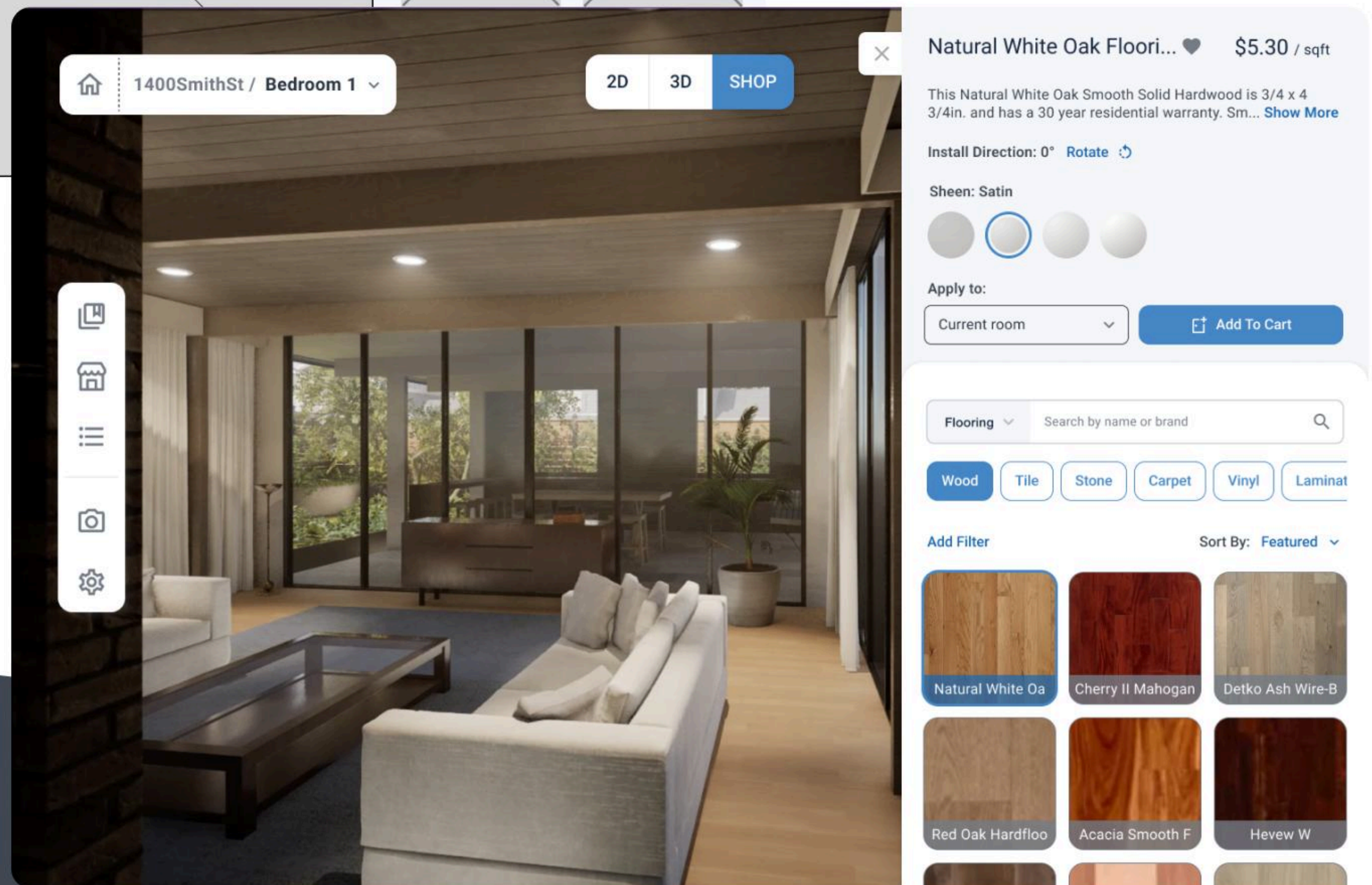


4 DESIGN

4.1 Wireframe / Prototype evolution

Low-Fidelity





4.2 Design System

Easy to use

Intuitive and fast turnaround of 3D model from 2D floor plans PDF

Reliable

Offering accurate takeoff calculations, that builders can access in real time

Modern

Simple, aesthetically pleasing and no-clutter visuals

Typeface

The primary font is Roboto. We use all of the fonts available in the Roboto family: Ultra Light, Thin, Light, Roman, Medium, Bold, Heavy and Black.

H1 Heading 1

Font weight Light | Font size 96 | Letter spacing 0.25 px

H2 Heading 2

Font weight Medium | Font size 60 | Letter spacing 0 px

H3 Heading 3

Font weight Medium | Font size 48 | Letter spacing -0.5 px

H4 Heading 4

Font weight Medium | Font size 32 | Letter spacing 0 px

H5 Heading 5

Font weight Regular | Font size 24 | Letter spacing 0 px

Body 1 (1rem)

Font weight Regular | Font size 16 | Letter spacing 0.25 px

Body 2

Font weight Regular | Font size 14 | Letter spacing 0.25 px

Subtitle

Font weight Medium | Font size 14 | Letter spacing 0.25 px

Button Large

Font weight Medium | Font size 20 | Letter spacing 0 px

Button Medium

Font weight Medium | Font size 16 | Letter spacing 0 px

Button Small

Font weight Bold | Font size 12 | Letter spacing 0 px

Caption

Font weight Regular | Font size 12 | Letter spacing 0.25 px

Colors

We use primary colors of the VIRTUALhaus brand as the main colors of the interface, complemented with greyscale.

Primary

#040F19	#16273F	#1A68B3	#4688C7	#BAD2E8	#E3EDF7
VIRTUALLY Black Backgrounds Text	VIRTUALLY Blue Backgrounds Text	Blueprint Blue Primary hover	Skylight Blue Brand elements Primary links	Blueprint Light Disabled states Light Background	Skylight Light Primary BG

Greyscale

#F6F7FA	#DEE1E5	#B5BDC7	#767D87	#414A54	#2A343F
Lightest Light Background	Light Disabled cards	Middle Disabled text	Dark Shadows	Text Middle Text titles	Text Text body

Text

Lines

Alerts



Success
Success text



Success Light
Success BG



Error
Error text



Error Light
Error BG



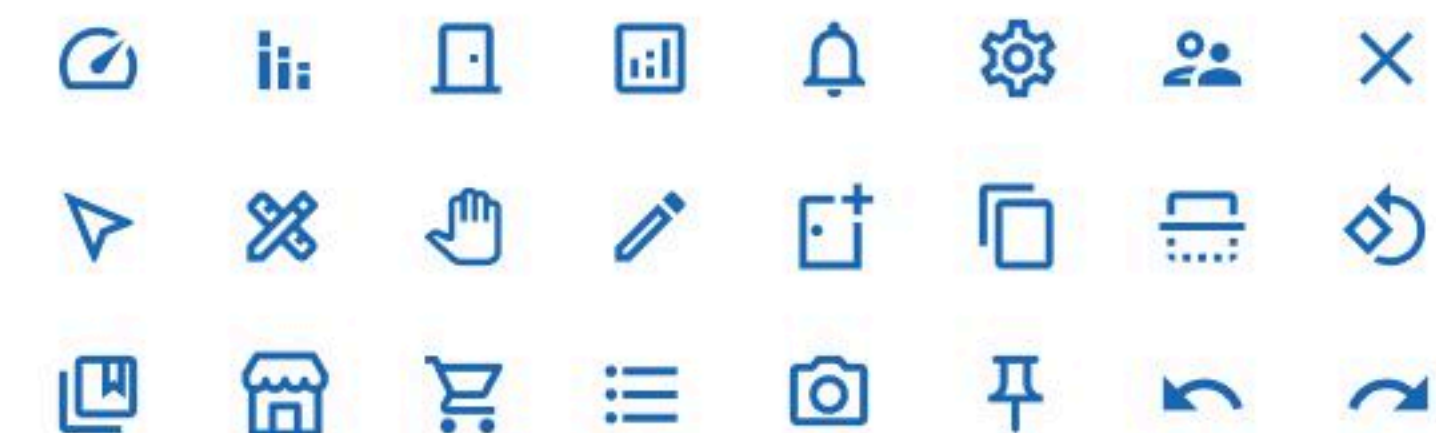
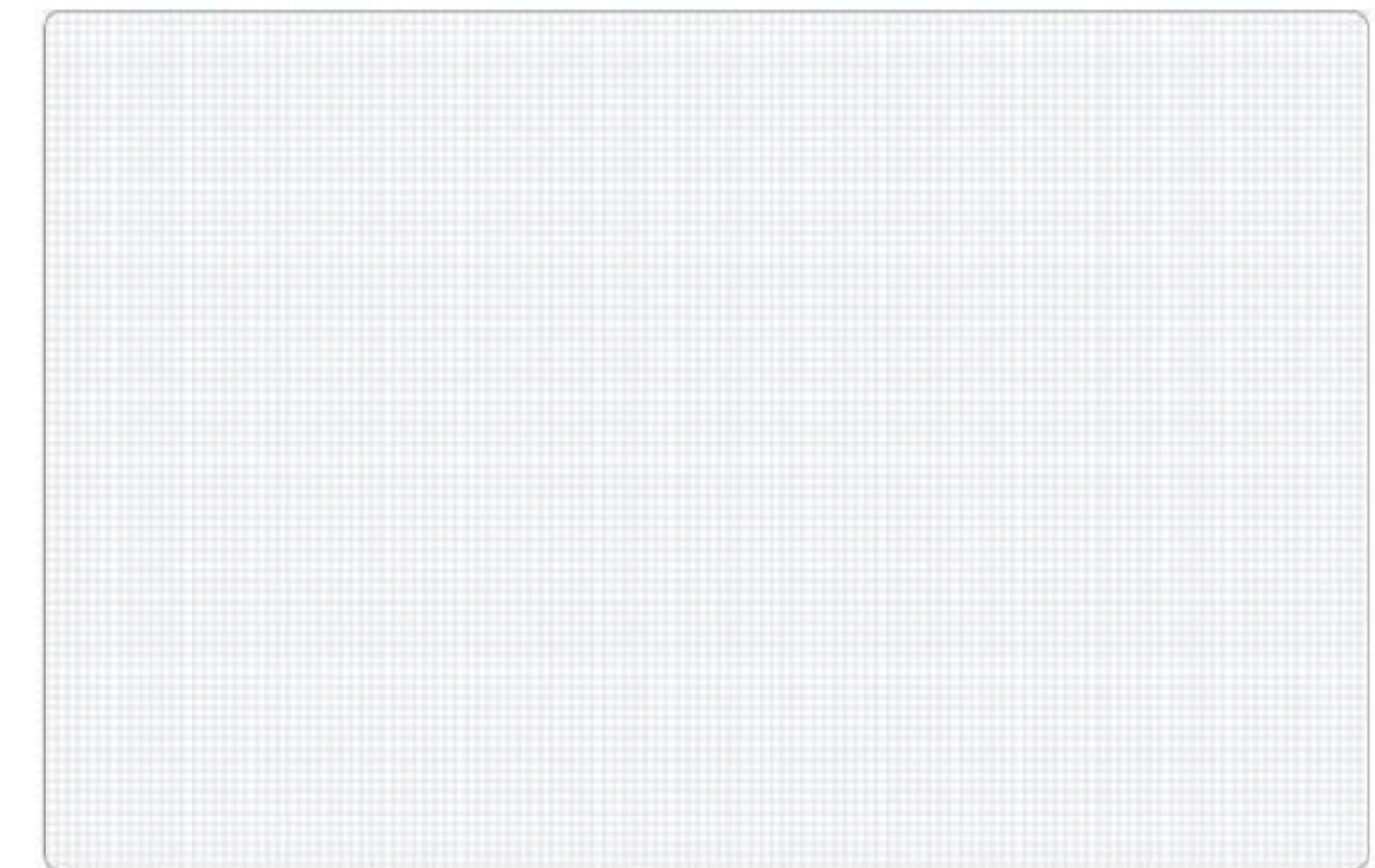
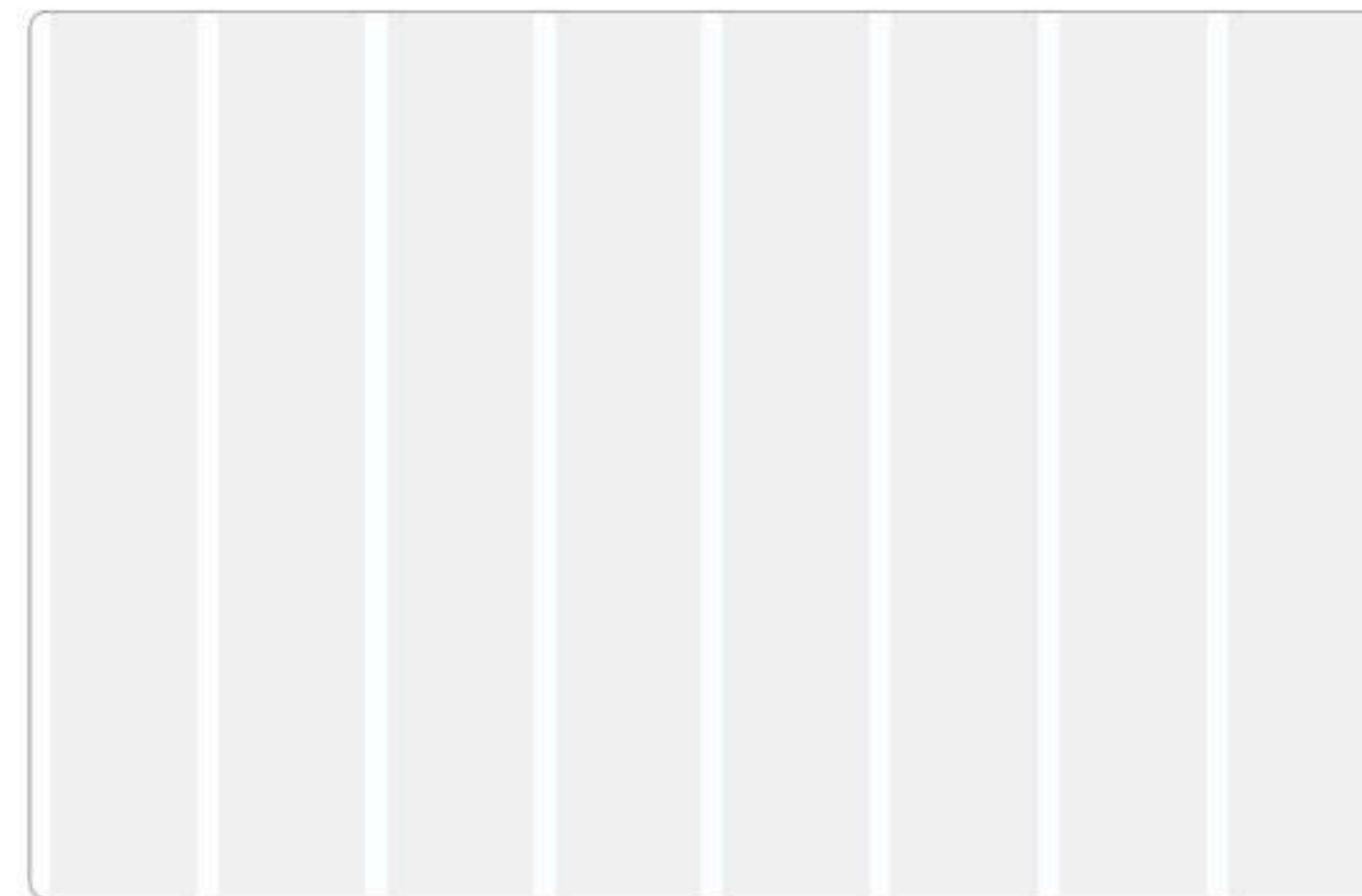
Warning
Warning text



Warning Light
Warning BG

Icons & Grids

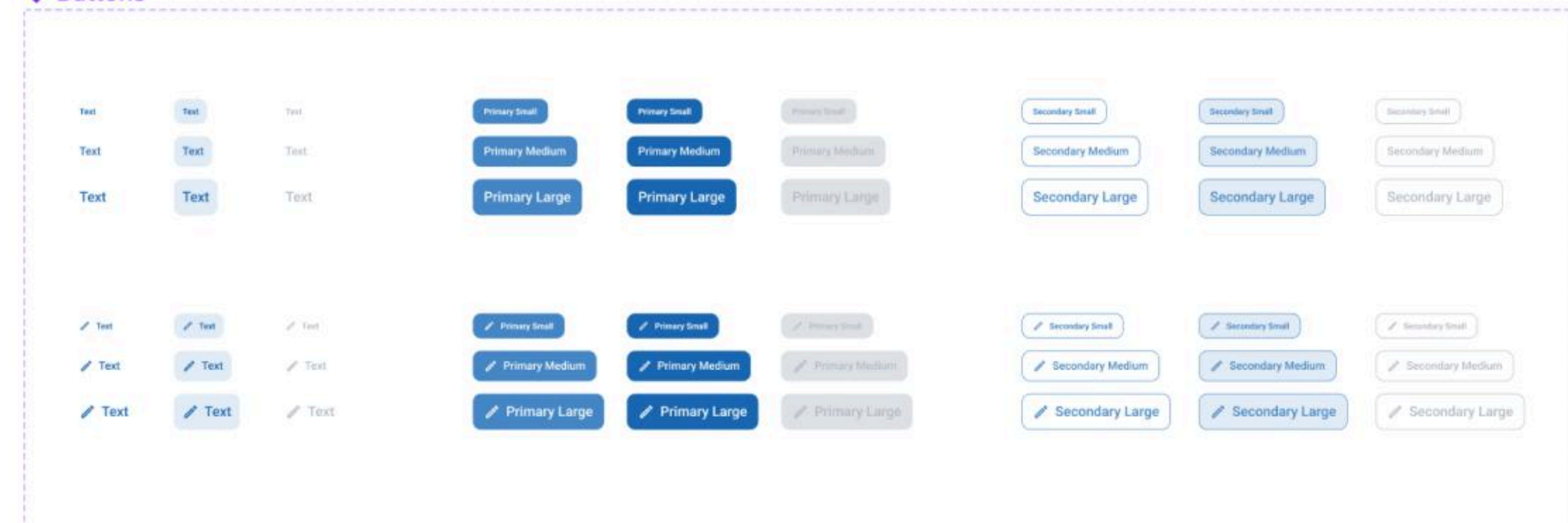
We use React Material outlined icons and some custom, an 8 column layout with a 24px margin and gutter, and a 12px grid.

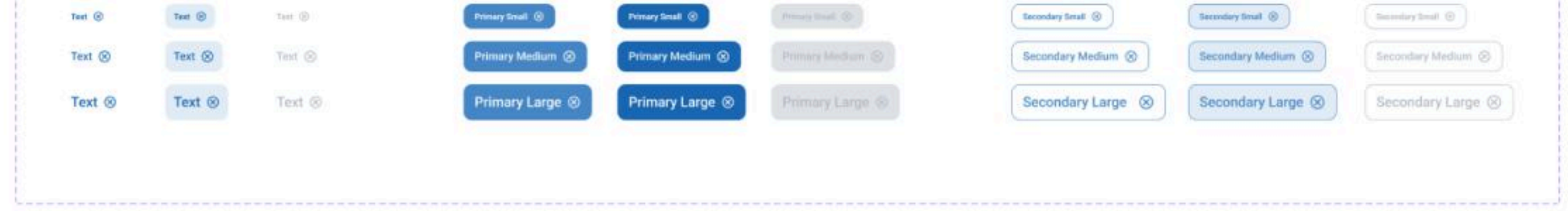


Components

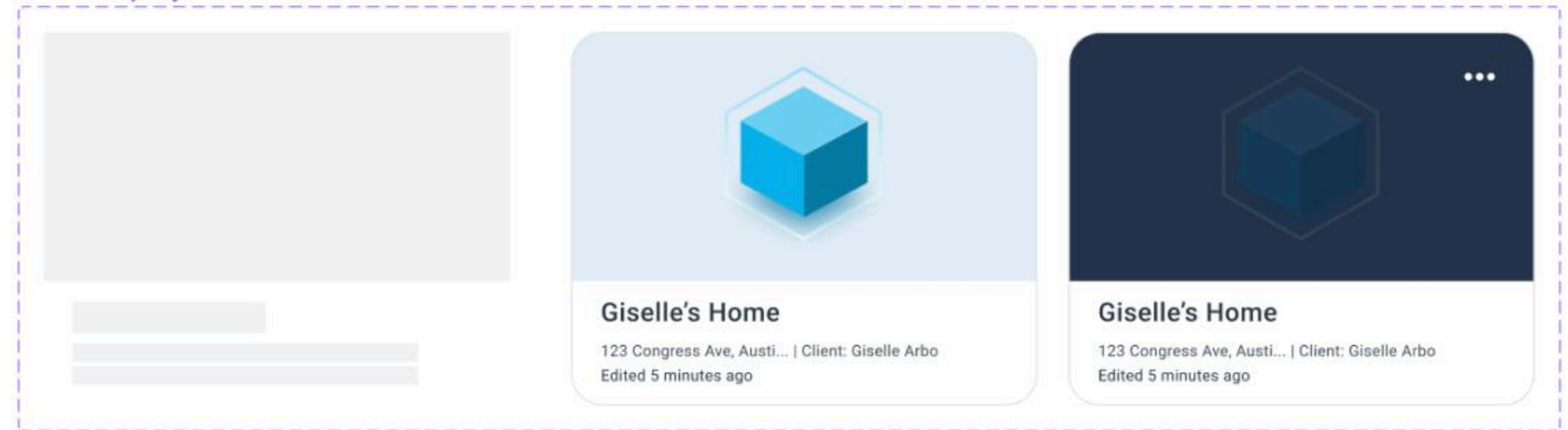
We use atomic Figma components that were created with auto layout and variants

Buttons

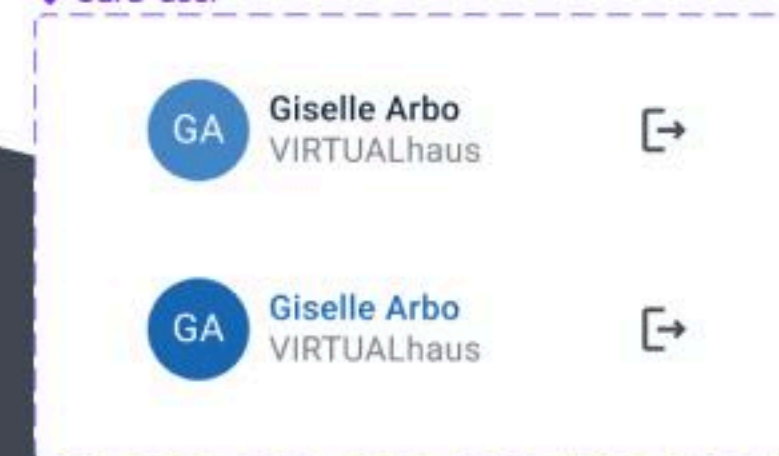




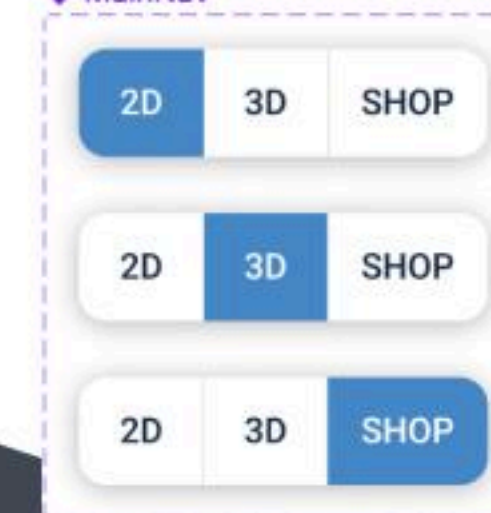
Card-project



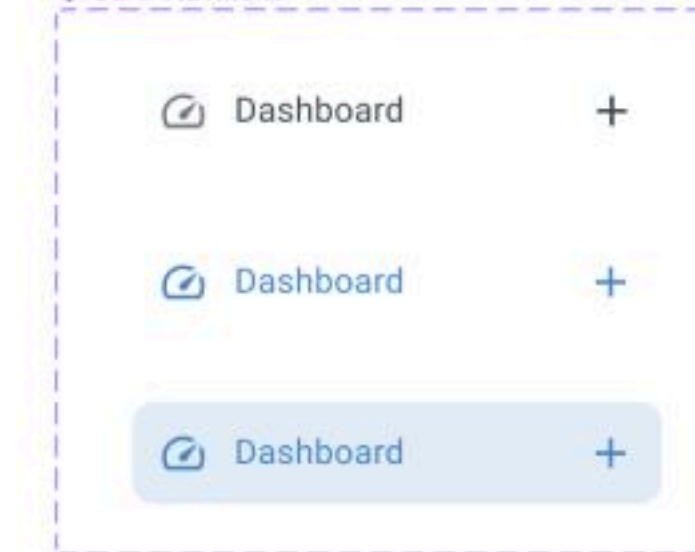
Card-user



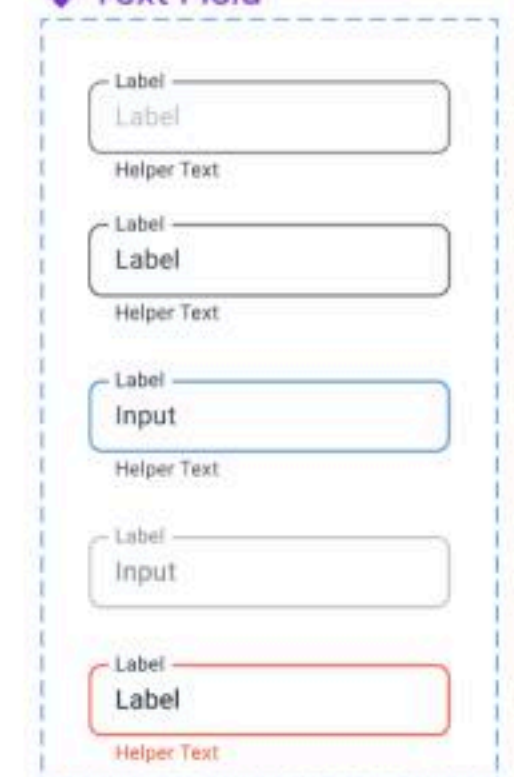
MainNav



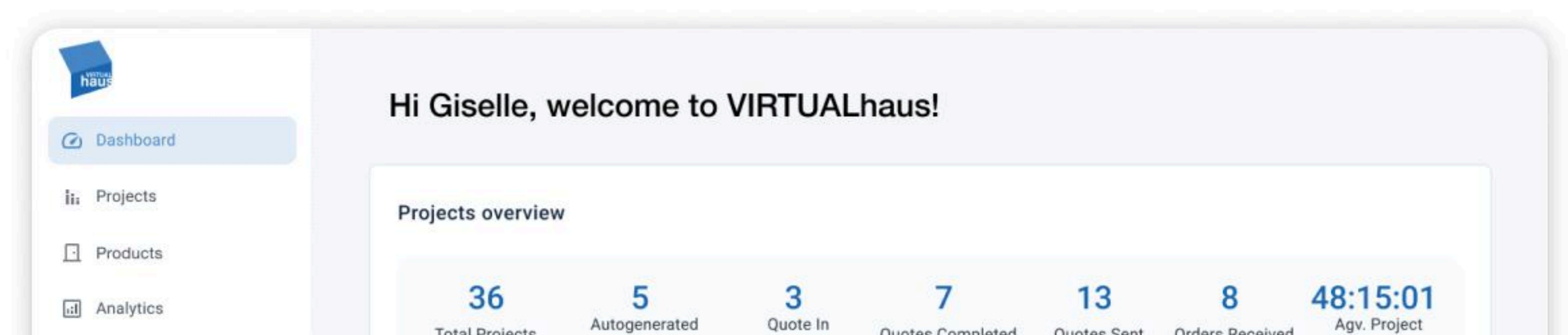
Card-Navmenu



Text Field



4.3 Portal Design



Notifications

Settings

Users & Permissions

Design

Create a 3D space

Launch Shop

Demo

Studio

Photo shoots

Recent Projects

View All



Giselle's Home

123 Congress Ave, Austi...

Edited 5 min ago



Single Famil..

No address added...

Notifications

View All

Autogeneration of you 3D Model is complete

Jul 9 at 10:59

Shane Indoben uploaded 2dplans.pdf

Jul 9 at 10:48

Chloe sent a 3D autogenerate request

Jul 9 at 10:13

Team Members

Abe Robertson

Chelsea Weber

Chloe Brown

Giselle Arbo (you)

Laura Dodds

Joe Timmons

Marc Kleinmann

Shane Inboden



Dashboard

Projects

Products

Analytics

Notifications

Settings

Users & Permissions

Projects

Search by project name, client or address

New Project

Show All Templates



Empty Project



NEW

Kitchen Template



NEW

Bathroom Template



Archer - 3Bed/2Bath



Westminster 4BR/4B...

Filter: All Projects

Sort: Last Viewed



Giselle's Home

123 Congress Ave, Austi... | Client: Giselle Arbo

Edited 5 minutes ago



Single Family Home 1

No addres added | No client added

Edited 1 day ago



Untitled-1

No addres added | No client added

Edited 1 day ago



Dashboard

Projects

Products

Analytics

Notifications

Settings

Users & Permissions

Analytics

Search by user ID, product or event name

Last 7 DaysLast 30 DaysAll Time

User BehaviorSales DataApp Stability

Seller: BFS

Top Brands

SKUs clicks (Impressions)



251

Active Products

1,435

SKUs Clicks

780

SKUs Added

Top Categories

	SKUs Clicks	No. of Searches
1. Baseboard	205	512
2. Casing	189	488
3. Counter	152	456
4. Kitchen Faucet	151	437
5. Kitchen Sink	147	331
6. Bathroom Sink	122	301
7. Flooring	99	260
8. Toilet	96	213
9. Crown Molding	84	175
10. Shower Set	63	148





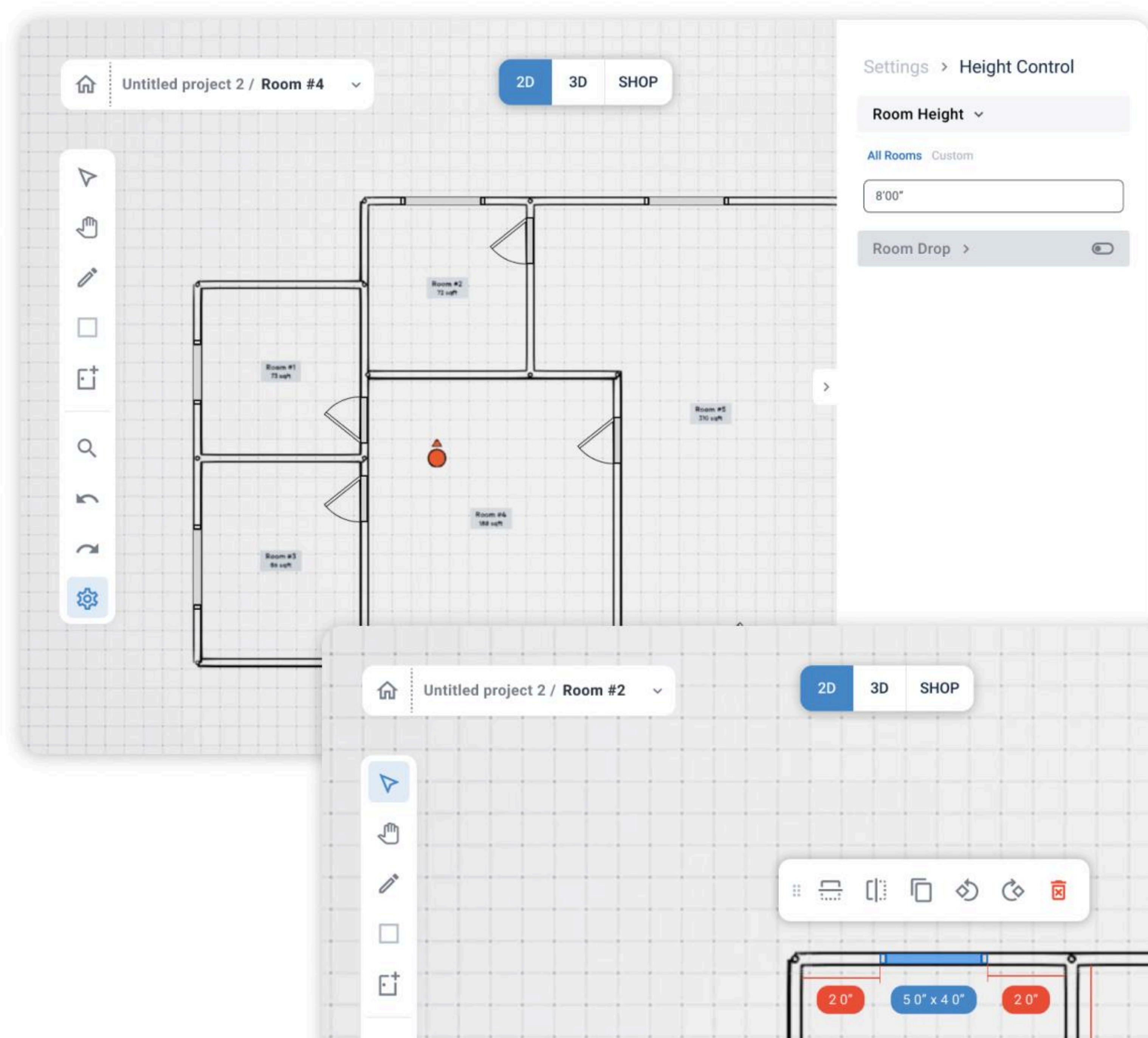
Giselle
VIRTUALhaus

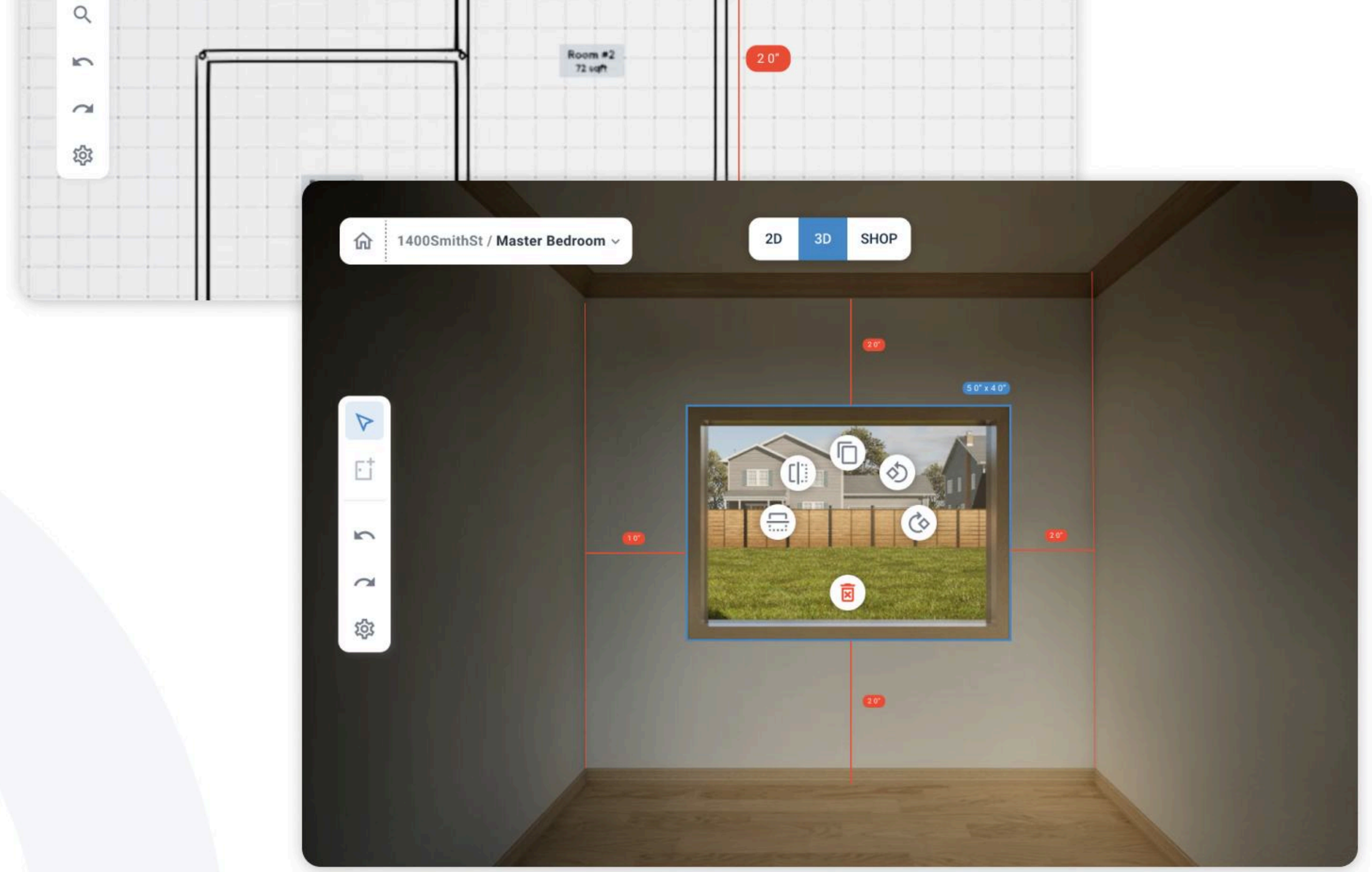
pw:a0a0aa
app:a0aaa00

Top Products

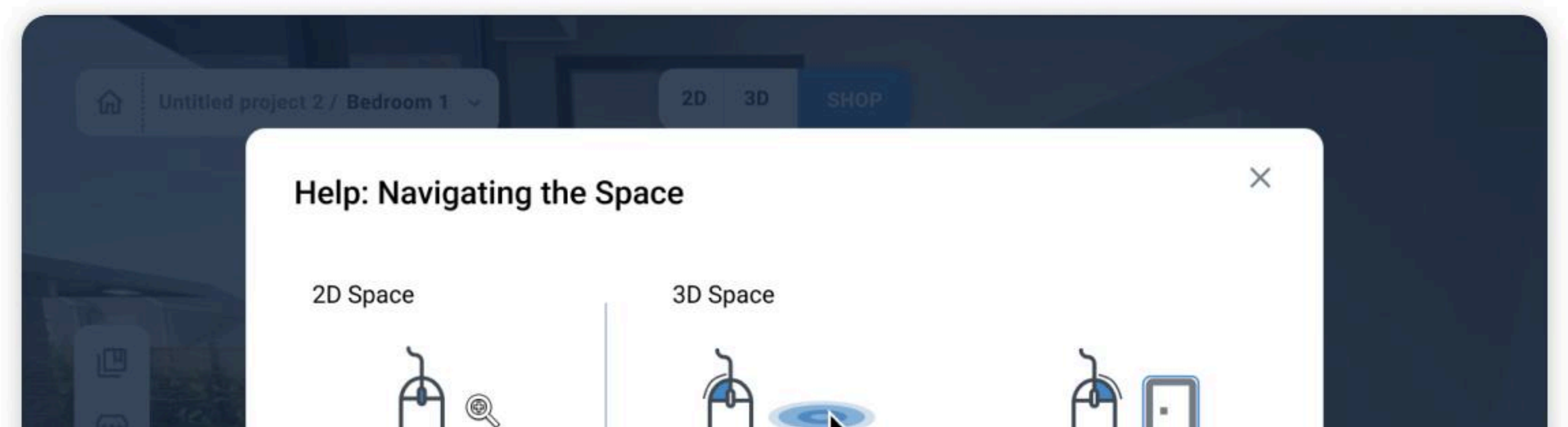
SKU Code	Product	Category	Brand	Seller	Impressions ↓
1. WERX-30194	100 Series Awning	Window	Andersen	BFS	526
2. BLNP-41563	400 Series Awning	Window	Andersen	BFS	397
3. NPYG-62396	400 Series Casement	Window	Andersen	BFS	318
4. BNVL-15028	5 Series Slider	Window	Andersen	BFS	255

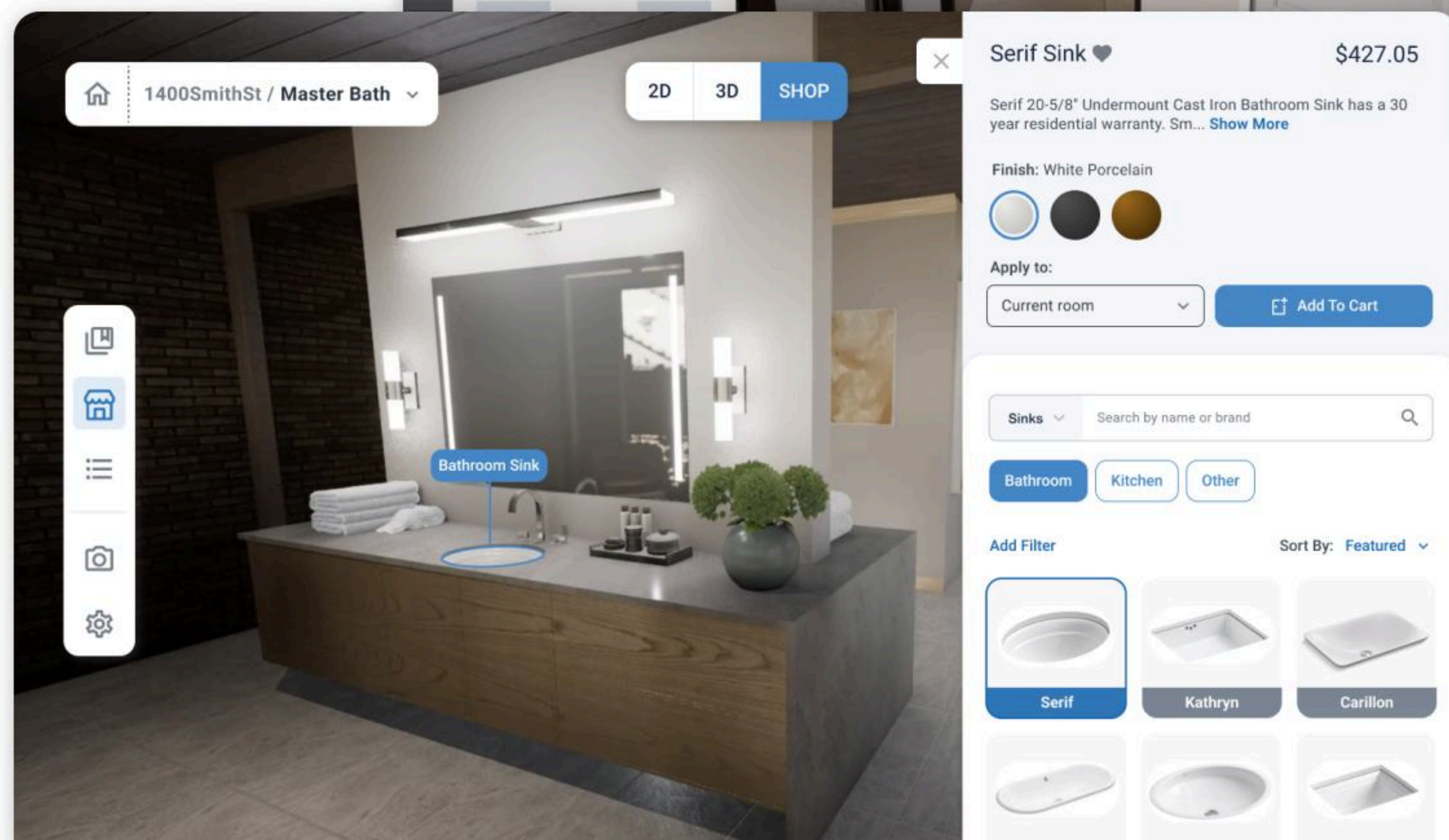
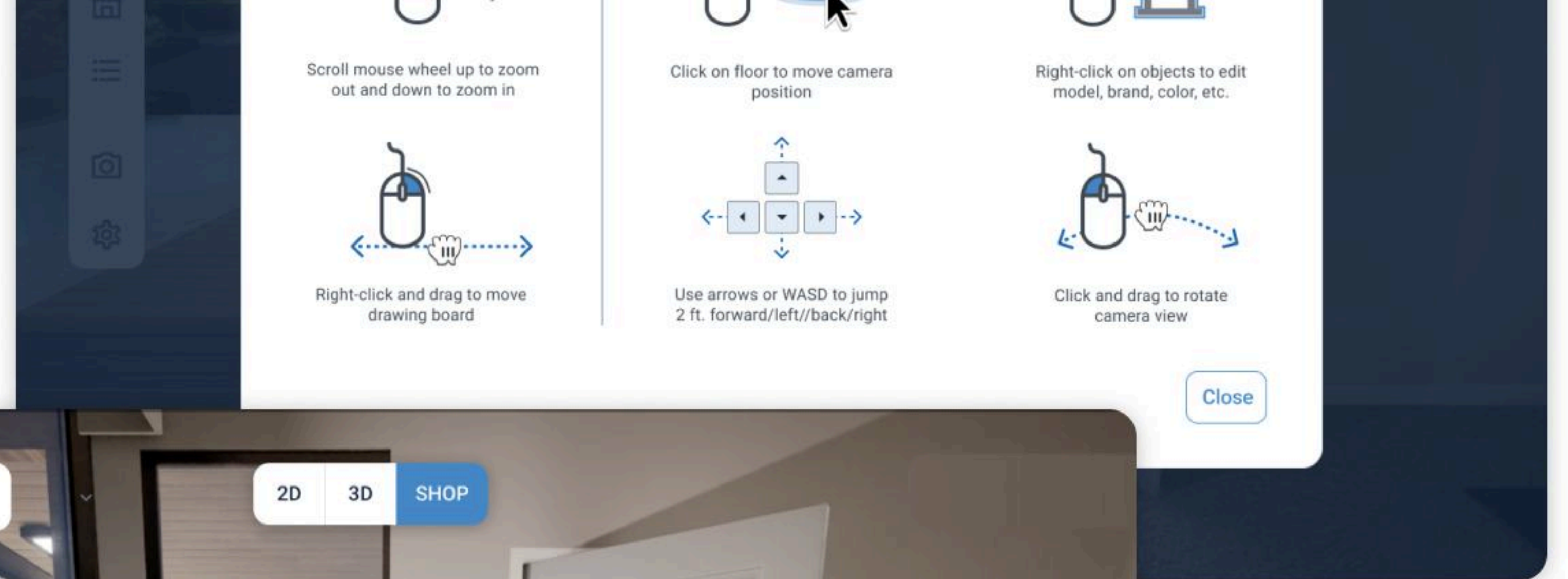
4.4 2D & 3D Space Design





4.5 3D Navigation and Shop Design





5 TESTING

5.1 Goals

We did usability tests to explore new features or evaluate existing ones as early as possible, trying to avoid developing features with design mistakes.

5.2 Usability test plan and script

All tests followed a plan and script to help speed the process. Part of one of the tests plans reads as followed:

“The testing goal is to gather qualitative behavioral data on 3D navigation. There will be at least 5 participants for best results. The test includes 2 specific tasks and 1 exploratory task.”

Script: Give a small intro explaining the App, telling participants that they are not being tested and that they should voice all their thoughts during the test.

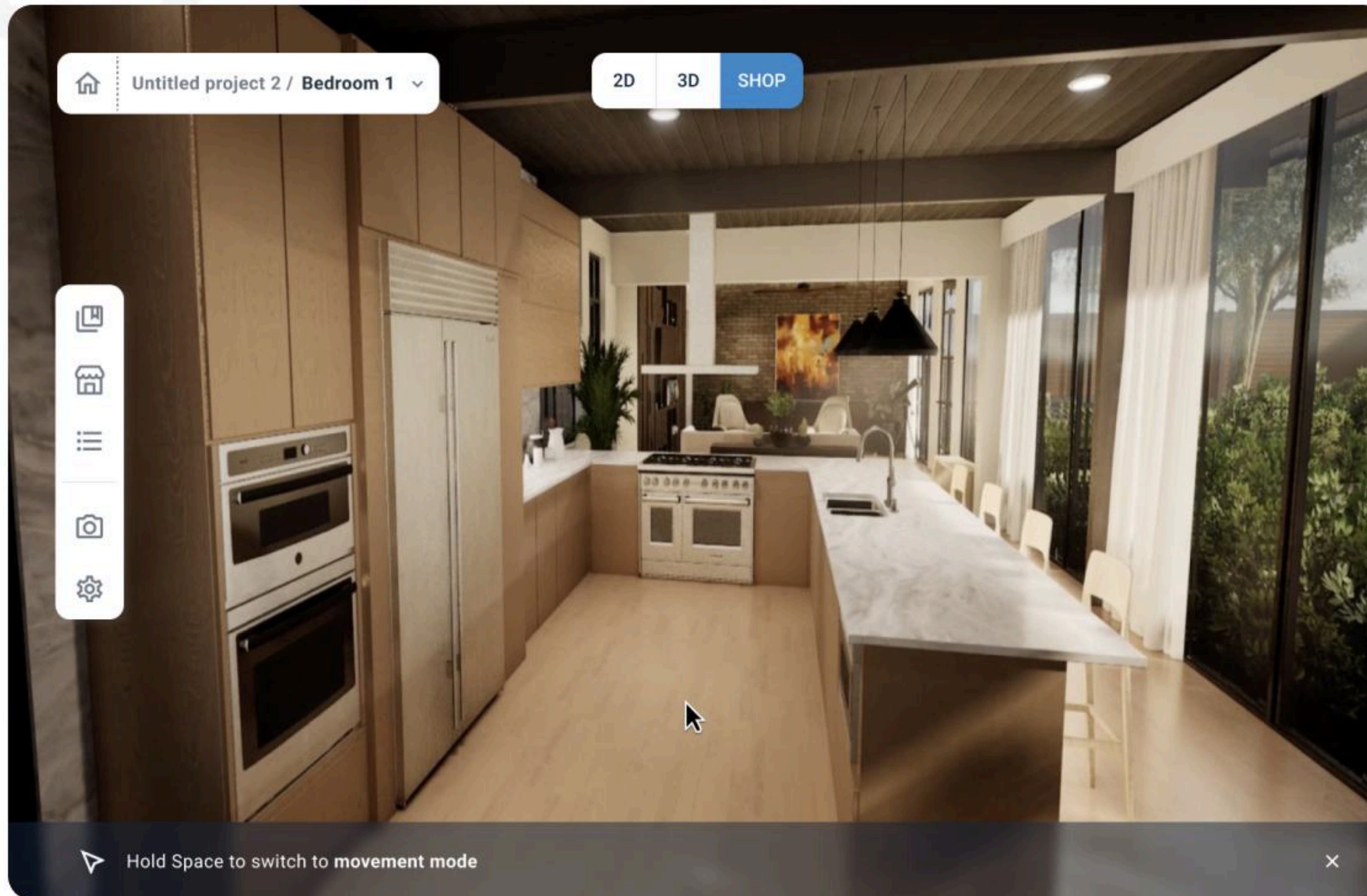
Task 1: “Find the kitchen sink and change to a different color.” The first task is an easy one where the 3D view starts facing away from the kitchen.

What we’re looking for: The user needs to orbit the camera to find the sink and select the product. Here we’re testing the orbit function and the product selection.

5.3 Tests and results

We held individual remote tests with at least 6 of our users. We recorded and took notes during all sessions to later analyze and put together the results reports.

Tested screen & navigation controls



6 out of 6 participants tried to orbit the camera with left-click.



5 out of 6 participants moved around the house with the Room Selector instead of walk tools.



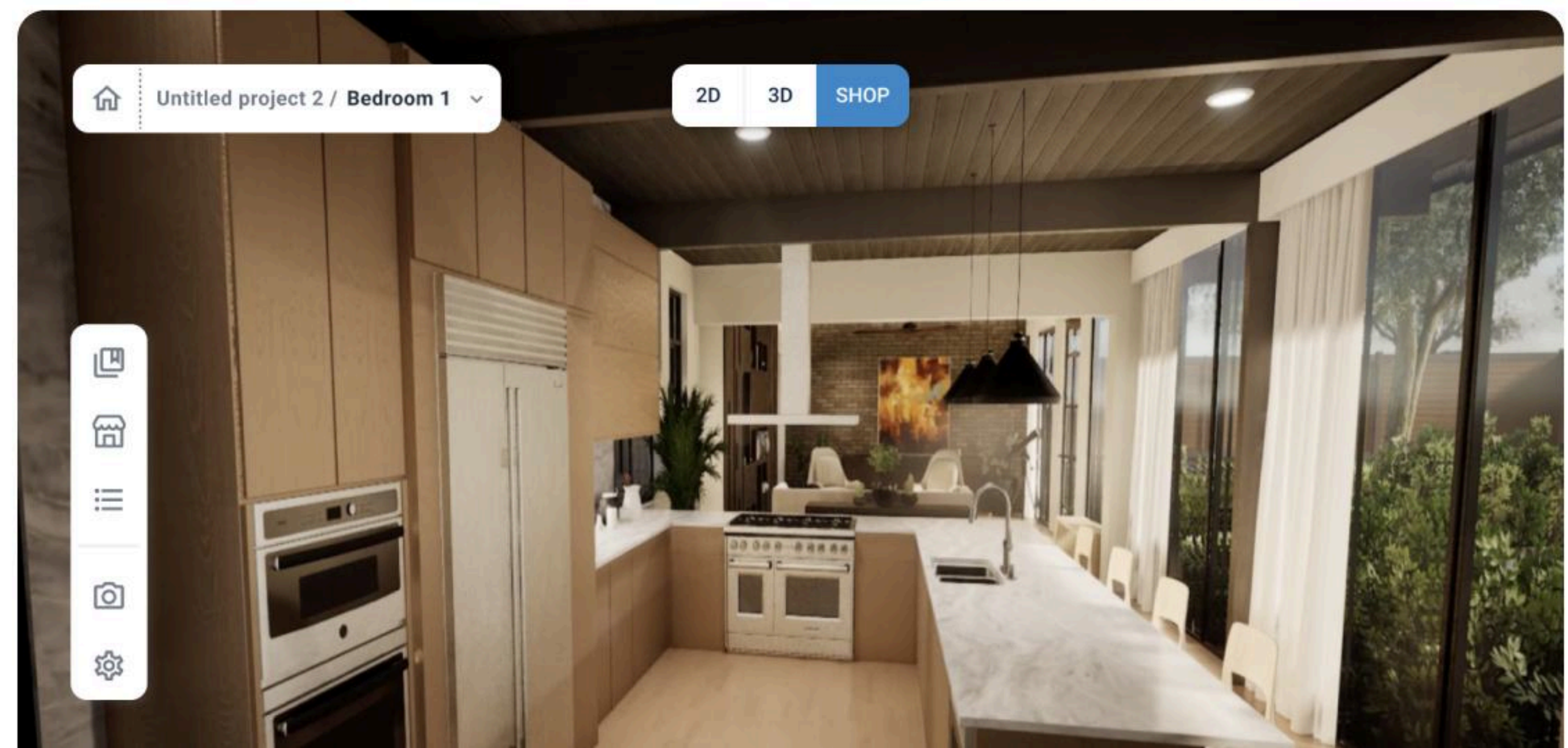
4 out of 6 participants expected right-click to open the menu to edit the product.

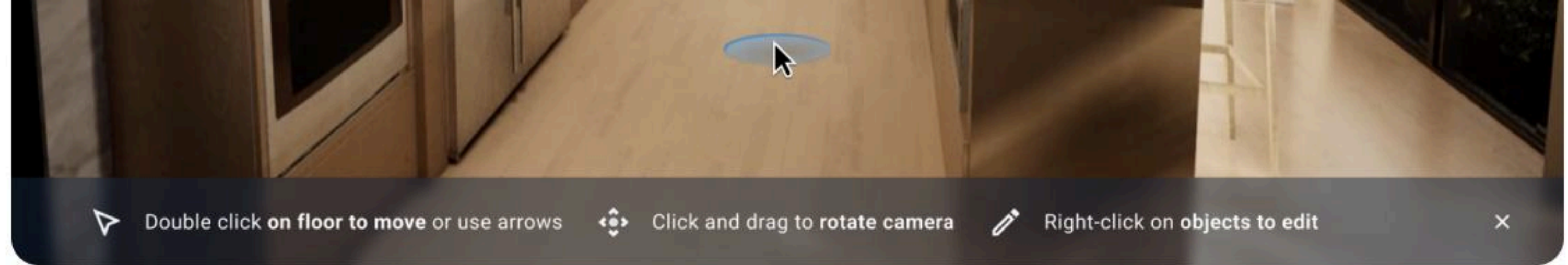


2 out of 6 participants only read the “Hold space..” tutorial after trying all possible keyboard and mouse functions.

Suggested screen & navigation controls

- ➡ Remove “Movement and Product Selection mode switching”.
- ➡ Show marker on floor and double click on floor to move position
- ➡ Holding a left-click orbits camera
- ➡ Hover on selectable objects highlights it and right-click opens edit/shop menu.





6 KEY TAKEAWAYS

6.1 Challenges

👤 Being a "UX/UI team of one" forced me to juggle many tasks at once, lacked a healthy dose of collaboration and meant that the responsibility for the success of the experience and interface was entirely in my hands.

6.2 Successes

- 👍 Optimized the 2D drawing and 3D editing tools, by adding custom cursors and drawing interactions.
- 👍 Solved a complex shopping experience by adding a hover effect on 3D shoppable products, highlighting and labeling the object.
- 👍 Reduced steps in shopping and adding to cart experience by simplifying a three-page panel into one page.
- 👍 Helped our user understand 3D navigation by adding a tutorial on launch.

Thank you for taking the time to read this far 🙏